

Predicting Maternal and Child Health Risk

A five-outcome risk cascade from two rounds of the Bangladesh Demographic and Health Survey

1. Problem Statement

Bangladesh has cut maternal and under-five mortality sharply over three decades, but the remaining burden is concentrated and unequal. Across the two survey rounds analysed here, 4.0% of live births end in neonatal death, 28.7% of measured children are stunted and 22.4% are underweight. Only 57.2% of recent births occur in a health facility — and that ranges from 34.0% in the poorest wealth quintile to 82.6% in the richest, a gap of 48.6 percentage points.

The difficulty is not that these risks are unknown; it is that they are identified too late. A community health worker meeting a pregnant woman for the first time has exactly what a household survey collects — age, parity, birth spacing, education, wealth, sanitation, reproductive history — and no laboratory or biomarker data at all.

Existing work on this data leaves four gaps. Studies predict one outcome in isolation, so a mother becomes a single number rather than a pathway. Complete-case restriction routinely shrinks a 30,000-record survey to two or three thousand rows. Leakage auditing is rarely reported and, judged by the performance figures, often absent — Naznin et al. (2025) report AUROC of 0.998 for under-five mortality from socio-demographic predictors alone, against a pooled accuracy of 68.9% across eleven DHS studies (Rao et al., 2025). And survey design is usually reduced to weighting, with clustering ignored, which understates every confidence interval.

Given only the information obtainable at a first antenatal contact, can one integrated system estimate risk across a connected panel of maternal and child health outcomes — with leakage-audited performance, design-correct national estimates, thresholds set for a realistic screening capacity, and behaviour validated both across survey rounds and across socio-economic strata?

2. Project Objectives

#	Objective	Completion criterion
01	Reproducible cleaning of two survey rounds	One identical pipeline for both rounds; decoding validated against a native DHS variable ($r = 0.999995$).
02	Preserve the nested universe structure	Four cohort views; no maternal-care or child-health outcome lost to a global missingness rule.
03	Design-correct national estimates	Ten indicators within 4 points of the published reports, computed per round with Taylor-linearised intervals.
04	Select targets on measured learnability	Fifteen candidates reduced to five, with stated evidence for each removal.
05	Eliminate leakage	Five leakage classes identified and blocked; before-and-after effect reported per target.
06	Benchmark current-generation models	Eight learners including CatBoost, RealMLP, TabM and TabPFN, under grouped cross-validation.
07	Operationally usable thresholds	Cost-optimal cut-points under an explicit false-negative penalty, capped at a 30% flag rate.
08	Validate transportability and equity	Temporal validation across rounds and subgroup performance by wealth, education and residence.

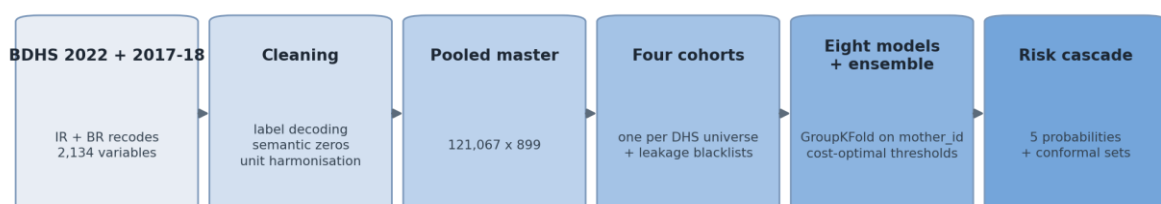


Figure 1 — The pipeline. Each stage writes artefacts the next reads from disk, so the deployment step can be exercised without the training code.

3. Literature Review

Eight studies frame this work; seven are from 2024 or later. Six use Bangladeshi DHS data directly, one pools six East African countries, and one is a meta-analysis establishing the realistic performance envelope.

Low birth weight. Sultana et al. (2025) applied six algorithms to 3,192 complete mother–child records from BDHS 2022. XGBoost led at 80% accuracy and AUC 0.761, with SHAP attributing most signal to gestational age and administrative division. The study is relevant to the newborn phase, but also illustrates the attrition problem addressed here: restricting to records with a measured birth weight discards the large majority of births.

Child undernutrition. Islam et al. (2024) modelled stunting, wasting and underweight on BDHS 2017-18 with roughly 7,800 children per outcome. XGBoost reached 81.7% accuracy at AUC 0.802 for stunting but only 76.2% at AUC 0.622 for wasting. That pairing — high accuracy with near-random discrimination — is exactly what a rare outcome produces when accuracy is the headline metric, and it motivates selecting on PR-AUC in this project.

Under-five mortality. Naznin et al. (2025) applied eight algorithms to 4,913 respondents from BDHS 2022, reporting random forest at 98.75% accuracy and 99.79% AUROC. The substantive determinants align with the wider literature, but the performance figures lie far outside what comparable studies obtain from socio-demographic predictors. Values of this kind usually indicate resampling before the train–test split, or post-outcome variables in the feature matrix; the paper is therefore treated as a cautionary reference and directly motivated this project's leakage audit.

Antenatal care. Salam, Islam and Karim (2025) predicted four-or-more antenatal visits from 5,128 women in BDHS 2022 using ten algorithms, with a random-forest variant best at 69.5% accuracy and AUC 0.734. The modest AUC is realistic for a behavioural outcome driven by access rather than physiology, and provides a credible external benchmark.

Continuum of care. Noor et al. (2025) predicted completion of the maternal continuum — four antenatal visits, skilled facility delivery, and postnatal care within 48 hours — among 5,128 women, with random forest at 82% accuracy and AUC 0.889. Only 25.3% of women completed the full continuum. This is the closest published analogue to the present work, treating care as a connected sequence; the difference is that this project predicts each stage separately and chains them, preserving interpretability at every step.

Performance envelope. A meta-analysis of eleven DHS-based studies (Rao et al., 2025) reports pooled accuracy of 68.92% for stunting, 84.39% for wasting and 73.60% for underweight, with high heterogeneity. Two implications shaped this project: the pooled figures establish what is realistically achievable, making any single study reporting near-perfect discrimination immediately suspect; and the concentration of published work on nutrition alone confirms that a connected maternal-and-child panel remains largely unaddressed.

Caesarean section. Zoha et al. (2025) predicted caesarean delivery from 2,490 BDHS 2022 records, with random forest at AUC 0.871 and a reported prevalence of 45.6% — far above the 10–15% the World Health Organization regards as clinically indicated. The outcome is highly predictable, but predictability here reflects health-system behaviour rather than obstetric need, which is why caesarean section was removed from this project's prediction panel and retained only as a descriptive equity indicator.

Multi-round pooling. Tadese et al. (2024) predicted incomplete immunization across six East African countries, pooling 27,691 children from DHS rounds between 2016 and 2021 under one harmonised definition; XGBoost led at 79.0% accuracy and AUC 0.86. Two features make it directly transferable: the multi-round strategy adopted here, and a predictor set that is almost entirely maternal and reproductive rather than clinical.

Modelling practice. Outside the DHS literature, TabArena (Erickson et al., 2025) benchmarks 16 models over 51 datasets and ranks CatBoost first among individual learners, while finding that a diverse post-hoc ensemble outperforms every single model. TabPFN (Hollmann et al., 2025), published in *Nature*, is a transformer pre-trained on synthetic tabular data that predicts in one forward pass. Both inform the model panel adopted in this project.

Identified gap. Across the reviewed work, no study predicts maternal and child outcomes as one connected panel; none pools BDHS rounds under a single harmonised pipeline; leakage auditing is rarely reported; survey design is usually reduced to weighting alone; and none reports whether performance differs across wealth quintiles — a striking omission in a literature whose subject is health inequity. This project addresses all five, and treats the lower, honest performance that results as the finding rather than a shortfall.

4. Dataset Identified

Field	Detail
Name	Bangladesh Demographic and Health Survey (BDHS) — 2022 (DHS-8) and 2017-18 (DHS-7)
Producer	NIPORT, Ministry of Health and Family Welfare, Government of Bangladesh, with ICF under The DHS Program (USAID-funded)
Recodes used	Individual Recode (one record per ever-married woman 15–49) and Birth Recode (one record per birth/pregnancy)
Format	Stata .DTA with embedded value labels; cleaned to Apache Parquet with a CSV mirror per cohort
Scale	2,134 raw variables; 899 retained. 121,067 pooled records — 73,239 from 2022 and 47,828 from 2017-18, across 1,346 clusters
Access	Free on registration; a short research-purpose statement is approved before download. Redistribution of raw files is not permitted, so this project ships code and derived artefacts.
Design	Stratified two-stage cluster sample; requires sampling weights (v005), primary sampling units (v021) and strata (v022) for valid inference

Source. Dataset catalogue:

<https://dhsprogram.com/data/available-datasets.cfm>

4.1 Why two rounds, and why four cohorts

Pooling raises the training sample by roughly 65%, which matters most for the rarest outcomes, and permits a trend comparison in which any difference between years cannot be an artefact of differing preparation. Two harmonisations were required: two administrative divisions were renamed between rounds, and cluster and mother identifiers are namespaced by survey, since cluster 1 in 2017-18 is a different village from cluster 1 in 2022.

DHS is a set of nested universes: mother-level variables for all records, birth history for live births, the maternity module only for recent births, anthropometry only for measured children. A column appears 93% missing when it belongs to an inner ring — it was never asked, not left unanswered — so any global missingness threshold deletes the inner rings first. The data is therefore sliced into four cohort views, with missingness judged only within a cohort.

Cohort	Targets carried	Records
cohort_births	neonatal death	112,546
cohort_mother	— features and equity analysis only	45,458
cohort_measured_child	stunting, severe stunting, underweight	11,954
cohort_recent_birth	facility delivery	10,773

4.2 The five prediction targets

Targets were chosen to be clinically actionable, measurable in both rounds, and spread across the phases of a pregnancy so the cascade tells a connected story: where she delivers, whether the baby survives, and how the child grows. All five are measured in both rounds, so every model trains on the pooled sample.

Target	Phase	Positive	Imbalance / cohort
facility_delivery	delivery	57.2%	0.7:1 — recent_birth
neonatal_death	newborn	4.0%	23.8:1 — births
stunted	child	28.7%	2.5:1 — measured_child
underweight_child	child	22.4%	3.5:1 — measured_child
severe_stunting	child	7.9%	11.6:1 — measured_child

An initial panel of fifteen outcomes was reduced to five after each was fitted and evaluated. Child fever (ROC-AUC 0.573), wasting (0.588) and diarrhoea (0.621) are two-week recall windows for acute infection, which is close to universally distributed — diarrhoea moves 0.4 percentage points across the entire wealth range. Caesarean section predicts well but records a clinician's decision rather than a patient's risk. Incomplete immunization appeared strong at 0.823 until the child's age revealed that every child under six months is “incomplete” by construction; restricted to the 12–23 month denominator DHS itself uses, it falls to 0.59.

4.3 Validation of the prepared data

Ten indicators were recomputed independently from the raw recodes and checked against the published reports per round; all ten fall within four percentage points, the largest deviation being 1.91. Stunting reproduces at 23.22% against a

published 23.6% for 2022 and 30.72% against 31.0% for 2017-18. Design effects average 2.19, so an analysis ignoring clustering would report confidence intervals about 1.48 times too narrow.

4.4 Ethical considerations and limitations

The DHS Program collects informed consent and releases only de-identified records; no participant can be re-identified from the variables used, and no GPS data was requested. Every record is retrospective, so covariates describe the mother at interview rather than at the time of an older pregnancy — all findings are associational. Neither round contains clinical or biomarker measurements, so pre-eclampsia, gestational diabetes, haemorrhage and sepsis have no ground truth and are not predicted. Outputs are population-level risk estimates intended to support triage by trained health workers; they are not a diagnosis and not a substitute for clinical assessment.

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